

Backups you can **trust**

because a restore is tested automatically, every day

Most teams



backup job ✓

restore: never tried



"it'll probably work"



This project



backup ✓ every hour

restore ✓ every day

6 s

restore + check, measured

0 orders missing

THE PROBLEM

How backups fail you 🙅

1 · Nightly backup



"success"

nobody reads the logs

2 · Months pass



never restored

"we'll test it someday"

3 · Bad day



DROP TABLE orders;

or: the disk dies

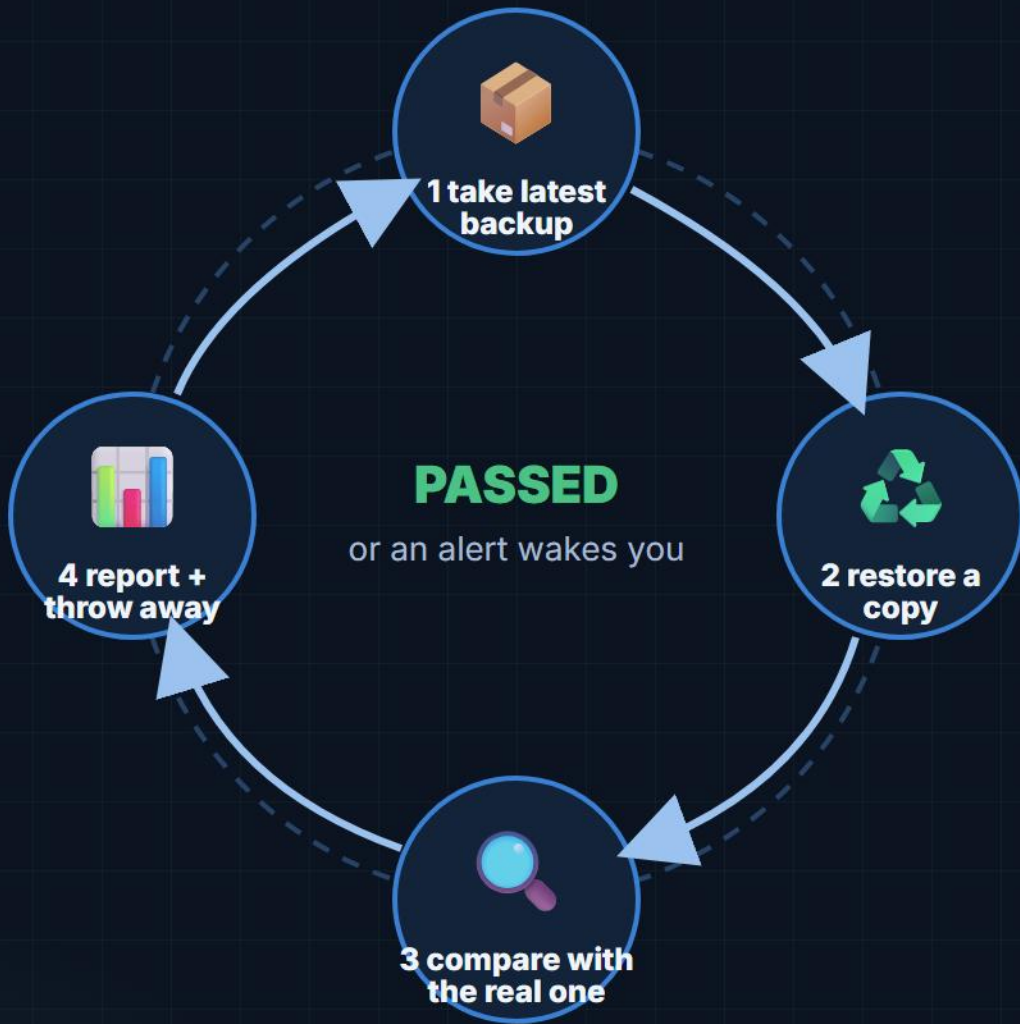
4 · The surprise



**corrupt · key lost ·
a day old · takes hours**

THE IDEA

A **fire drill** for your data, every night

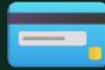


WHEN TO USE IT

Great for... / skip it when...



PostgreSQL you run yourself (VM, server)



orders, payments, user data



audits ask for RTO / RPO proof



"undo" a bad query or migration

Skip it when...



a managed database already does backups
(but still run a restore drill against it!)



throwaway test databases



caches you can rebuild from elsewhere

HOW A TEAM USES IT

Set up once, then it proves itself

1



scripts/up.sh

secrets, certificates, first backup: 68 s

2



Backups run themselves

changes every ≤ 60 s · hourly · daily · weekly

3



Nightly restore drill

+ every file checked against its checksum

4



Glance at the dashboard

PASSED · restore time · data loss · 0 missing

5

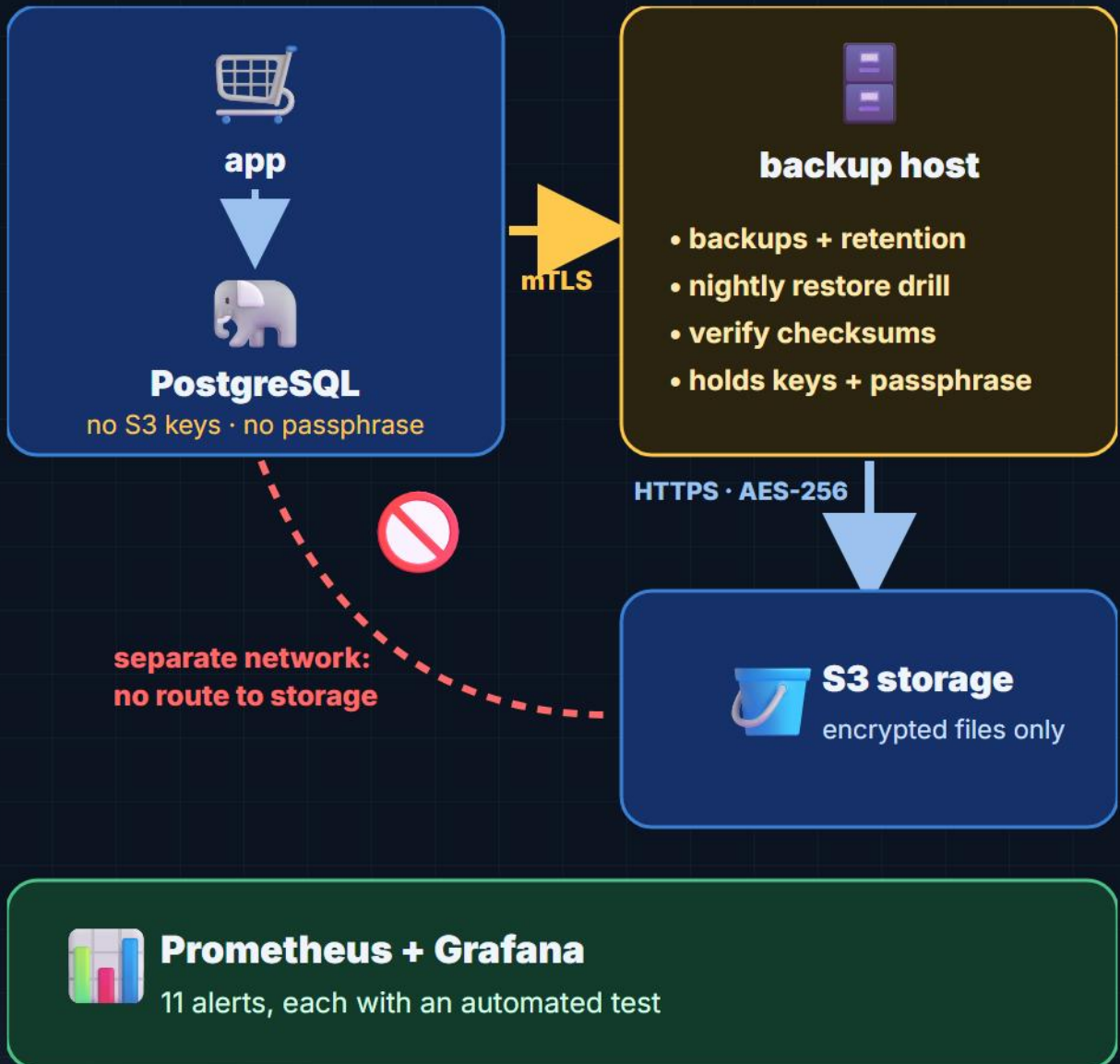


Bad day? One command

back to the second before the mistake

ARCHITECTURE

The database can't touch its backups



HOW IT WORKS INSIDE

Every minute, hour, day and week



every ≤ 60 s

new changes (WAL) archived → undo to any second



every hour

incremental backup: only changed blocks



every day

differential backup · verify every file



every night 04:00

restore drill: copy → start → compare → amcheck



every week

full backup · keep the last 2 fulls



anything fails?

alert: drill failed · backup old · WAL stuck · file damaged

I broke it on purpose 6 ways

💣 DROP TABLE orders

10 s

123 / 123 orders back
to the second before

💾 database disk lost

16 s

rebuilt from S3
every archived order back

📶 storage down

0 lost

database kept working
alert, then caught up in 35 s

🦋 4 KB of bit rot

caught

verify + drill failed
2 alerts, before you need it

🔑 wrong passphrase

alert

nothing restorable
→ store the key safely!

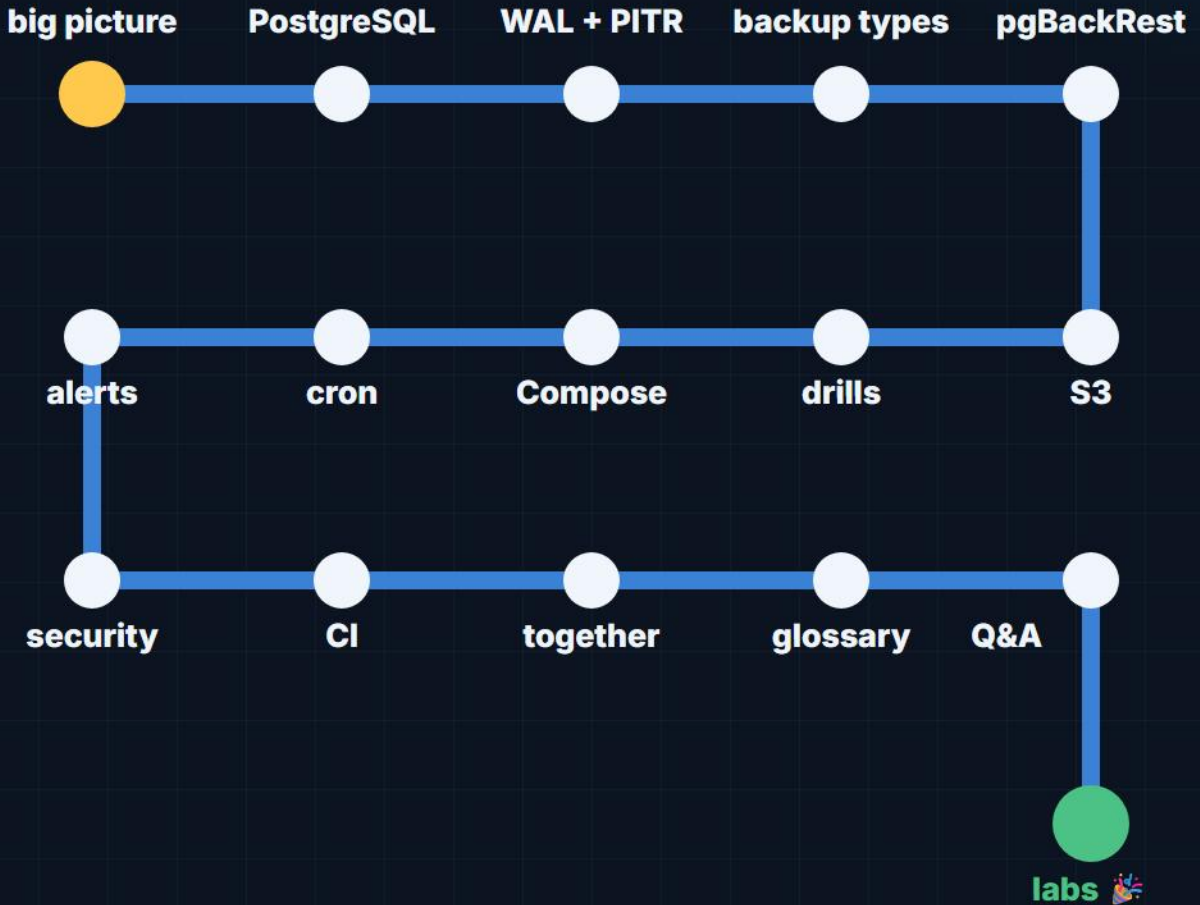
🧯 nightly drill

6 s

0 missing · no corruption
data loss ≤ ~1 minute

FREE STUDY MATERIAL

Learn every piece from zero



49

page PDF

10

hands-on labs

25

interview Q&A

TRY IT YOURSELF

Only Docker needed · ~15 min



```
$ scripts/up.sh
```

stack + first backup



```
$ scripts/drill.sh
```



PASSED in 6 s



```
$ scripts/pitr.sh "14:05"
```



undo



```
$ scripts/recover-lost-db.sh
```



```
$ scripts/status.sh
```



```
$ scripts/e2e.sh
```

all



YOUR TURN

When did you last **restore** a backup?



Code + study guide

github.com/sufyanahmadkamboh



All my projects

sufyanahmadkamboh.github.io

Tell me in the comments 🙌

this week? this year? never?